

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Designed and shipped to production a self-service analytics agent over 67 BigQuery tables with sub-agent orchestration and context-isolated validation (Context Researcher → SQL Drafter → Validator → Devil's Advocate); 8 deterministic SQL rules and a 23-case golden-query evaluation suite for reliability. Approved by Sr. Director and product leadership and currently the only AI skill in active use by business teams on Data Ventures — a natural-language front door to business systems.
- Built and shipped to production an autonomous Jira ticket-resolution agent with agentic RAG: 6-gate triage, MCP-driven context gathering, FAISS + BM25 retrieval over 11,000 historical SQL queries, and self-healing execution with a 10-retry budget. Closed 400+ data requests across 8 categories over 15 months and cut routine turnaround from 30–60 minutes to under 10 minutes per request.
- Built a registry-governed semantic layer over BigQuery that boxes the LLM into semantic parsing — one call, strict JSON intent, never SQL — with deterministic validation, value canonicalization, and a deterministic SQL compiler, making answers reproducible and the only probabilistic step regression-testable.
- Built a hybrid orchestrator on top of the agent that lets Product self-serve requests they previously filed as tickets; routes questions through a semantic-layer skill and the context-engineered agent, reasons about disagreements, and feeds discrepancies back to refine definitions over time (agent self-improvement loop).
- Built and deployed to production a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; decision logs and state persist to BigQuery for end-to-end observability (durable agent memory, no LLM-memory reliance).
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- RAG-to-proposal generator deployed on AWS EC2; cut proposal time ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit front-end + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

AI / agent systems: agent orchestration, sub-agent orchestration, agent memory / durable state, context engineering, prompt engineering, RAG (embeddings + FAISS + BM25 + RRF), MCP server design, LLM self-reflection / refinement loops, evaluation suites (golden sets, human-in-the-loop, accuracy/latency), multi-model routing across providers

Backend / production: Python, FastAPI, Cloud Run, REST APIs, CI/CD (pytest, version control), headless cron agents, event-driven architectures, decision logging / observability, JSONL state stores, Playwright SSO

Data / SQL: BigQuery, dbt, SQL (complex query design + optimization), knowledge/semantic graphs (registry-governed semantic layer), PySpark, Airflow, Databricks, MySQL

LLMs: Claude (Opus / Sonnet / Haiku), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

Viz: Power BI (DAX, XMLA TMSL), Tableau, Streamlit, Matplotlib

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL